

Sharma_DSC520Week11.3FinalProjectStep3

2025-02-27

```
churn <- read.csv('Telco-Customer-Churn.csv')
str(churn)

## 'data.frame': 7043 obs. of 21 variables:
## $ customerID : chr "7590-VHVEG" "5575-GNVDE" "3668-QPYBK" "7795-CFOCW" ...
## $ gender : chr "Female" "Male" "Male" "Male" ...
## $ SeniorCitizen : int 0 0 0 0 0 0 0 0 0 ...
## $ Partner : chr "Yes" "No" "No" "No" ...
## $ Dependents : chr "No" "No" "No" "No" ...
## $ tenure : int 1 34 2 45 2 8 22 10 28 62 ...
## $ PhoneService : chr "No" "Yes" "Yes" "No" ...
## $ MultipleLines : chr "No phone service" "No" "No" "No phone service" ...
## $ InternetService : chr "DSL" "DSL" "DSL" "DSL" ...
## $ OnlineSecurity : chr "No" "Yes" "Yes" "Yes" ...
## $ OnlineBackup : chr "Yes" "No" "Yes" "No" ...
## $ DeviceProtection: chr "No" "Yes" "No" "Yes" ...
## $ TechSupport : chr "No" "No" "No" "Yes" ...
## $ StreamingTV : chr "No" "No" "No" "No" ...
## $ StreamingMovies : chr "No" "No" "No" "No" ...
## $ Contract : chr "Month-to-month" "One year" "Month-to-month" "One year" ...
## $ PaperlessBilling: chr "Yes" "No" "Yes" "No" ...
## $ PaymentMethod : chr "Electronic check" "Mailed check" "Mailed check" "Bank transfer (automatic)" ...
## $ MonthlyCharges : num 29.9 57 53.9 42.3 70.7 ...
## $ TotalCharges : num 29.9 1889.5 108.2 1840.8 151.7 ...
## $ Churn : chr "No" "No" "Yes" "No" ...

sapply(churn, function(x) sum(is.na(x)))

## customerID gender SeniorCitizen Partner
## 0 0 0 0
## Dependents tenure PhoneService MultipleLines
## 0 0 0 0
## InternetService OnlineSecurity OnlineBackup DeviceProtection
## 0 0 0 0
## TechSupport StreamingTV StreamingMovies Contract
## 0 0 0 0
## PaperlessBilling PaymentMethod MonthlyCharges TotalCharges
## 0 0 0 11
## Churn
## 0

churn <- churn[complete.cases(churn), ]

cols_recode1 <- c(10:15)
for(i in 1:ncol(churn[,cols_recode1]))
{
```

```

churn[,cols_recode1][,i] <- as.factor(mapvalues
  (churn[,cols_recode1][,i], from=c("No internet service"),to=c("No")))
}

churn$MultipleLines <- as.factor(mapvalues(churn$MultipleLines,
  from=c("No phone service"),to=c("No")))

min(churn$tenure); max(churn$tenure)

## [1] 1
## [1] 72

group_tenure <- function(tenure)
{
  if (tenure >= 0 & tenure <= 12)
  {
    return('0-12 Month')
  }
  else if(tenure > 12 & tenure <= 24)
  {
    return('12-24 Month')
  }
  else if (tenure > 24 & tenure <= 48)
  {
    return('24-48 Month')
  }
  else if (tenure > 48 & tenure <=60)
  {
    return('48-60 Month')
  }
  else if (tenure > 60)
  {
    return('> 60 Month')
  }
}

churn$tenure_group <- sapply(churn$tenure,group_tenure)
churn$tenure_group <- as.factor(churn$tenure_group)

```

Data Cleaning

customerID has no significance so we are dropping this column

```

churn <- subset(churn, select = -customerID)
head(churn)

```

```

##   gender SeniorCitizen Partner Dependents tenure PhoneService MultipleLines
## 1 Female           0     Yes         No       1           No           No
## 2  Male           0     No          No      34           Yes           No
## 3  Male           0     No          No       2           Yes           No
## 4  Male           0     No          No      45           No           No
## 5 Female           0     No          No       2           Yes           No
## 6 Female           0     No          No       8           Yes           Yes

```

```
##      InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport
## 1          DSL          No          Yes          No          No
## 2          DSL          Yes          No          Yes          No
## 3          DSL          Yes          Yes          No          No
## 4          DSL          Yes          No          Yes          Yes
## 5      Fiber optic          No          No          No          No
## 6      Fiber optic          No          No          Yes          No
##      StreamingTV StreamingMovies      Contract PaperlessBilling
## 1          No          No Month-to-month          Yes
## 2          No          No      One year          No
## 3          No          No Month-to-month          Yes
## 4          No          No      One year          No
## 5          No          No Month-to-month          Yes
## 6          Yes          Yes Month-to-month          Yes
##      PaymentMethod MonthlyCharges TotalCharges Churn tenure_group
## 1      Electronic check          29.85          29.85    No 0-12 Month
## 2          Mailed check          56.95          1889.50    No 24-48 Month
## 3          Mailed check          53.85          108.15   Yes 0-12 Month
## 4 Bank transfer (automatic)          42.30          1840.75    No 24-48 Month
## 5      Electronic check          70.70          151.65   Yes 0-12 Month
## 6      Electronic check          99.65          820.50   Yes 0-12 Month
```

```
# Handling Missing, Duplicate Data and null data
sum(duplicated(churn))
```

```
## [1] 22
```

```
# There are 17 Duplicate rows & 11 missing values in TotalCharges Column.
# The number of Duplicate rows & missing values is relatively small compared
# to the size of the dataset. Therefore, I have decided to remove these missing
# values from our analysis.
```

```
datc <- churn %>%
  na.omit(churn) %>%
  distinct()
sum(duplicated(datc))
```

```
## [1] 0
```

Checking Data structure

```
summary(churn)
```

```
##      gender      SeniorCitizen      Partner      Dependents
## Length:7032      Min.      :0.0000      Length:7032      Length:7032
## Class :character 1st Qu.:0.0000      Class :character  Class :character
## Mode  :character Median :0.0000      Mode  :character  Mode  :character
##                      Mean  :0.1624
##                      3rd Qu.:0.0000
##                      Max.  :1.0000
##      tenure      PhoneService      MultipleLines      InternetService
## Min.      : 1.00      Length:7032      No :4065      Length:7032
## 1st Qu.: 9.00      Class :character  Yes:2967      Class :character
## Median :29.00      Mode  :character                      Mode  :character
## Mean      :32.42
```

```
## 3rd Qu.:55.00
## Max. :72.00
## OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV
## No :5017 No :4607 No :4614 No :4992 No :4329
## Yes:2015 Yes:2425 Yes:2418 Yes:2040 Yes:2703
##
##
##
## StreamingMovies Contract PaperlessBilling PaymentMethod
## No :4301 Length:7032 Length:7032 Length:7032
## Yes:2731 Class :character Class :character Class :character
## Mode :character Mode :character Mode :character
##
##
##
## MonthlyCharges TotalCharges Churn tenure_group
## Min. : 18.25 Min. : 18.8 Length:7032 > 60 Month :1407
## 1st Qu.: 35.59 1st Qu.: 401.4 Class :character 0-12 Month :2175
## Median : 70.35 Median :1397.5 Mode :character 12-24 Month:1024
## Mean : 64.80 Mean :2283.3 24-48 Month:1594
## 3rd Qu.: 89.86 3rd Qu.:3794.7 48-60 Month: 832
## Max. :118.75 Max. :8684.8
```

The “SeniorCitizen” variable in the dataset is currently coded as a binary values of “0” or “1”. This coding scheme is not intuitive and can be difficult. To make our analysis easier and more understandable, we can recode this intuitive labeling scheme, such as “yes” and “no”.

The “MultipleLines” variable is related to the “PhoneService” variable. If “PhoneService” is set to “No”, then “MultipleLines” is also set to “No”. To simplify our graphics and modeling, we can change the “No phone service” “MultipleLines” variable to “No”. Same goes with “No internet service”

```
datrecode <- churn %>%
mutate(SeniorCitizen = if_else(SeniorCitizen == 0, "No", "Yes")) %>%
mutate(MultipleLines = if_else(MultipleLines == "No phone service", "No",
                             MultipleLines)) %>%
mutate(across(starts_with("Online") | starts_with("Device") |
               starts_with("Tech") | starts_with("Streaming"),
               ~if_else(. == "No internet service", "No", .)))
head(datrecode)
```

```
## gender SeniorCitizen Partner Dependents tenure PhoneService MultipleLines
## 1 Female No Yes No 1 No No
## 2 Male No No No 34 Yes No
## 3 Male No No No 2 Yes No
## 4 Male No No No 45 No No
## 5 Female No No No 2 Yes No
## 6 Female No No No 8 Yes Yes
```

	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport
## 1	DSL	No	Yes	No	No
## 2	DSL	Yes	No	Yes	No
## 3	DSL	Yes	Yes	No	No
## 4	DSL	Yes	No	Yes	Yes
## 5	Fiber optic	No	No	No	No
## 6	Fiber optic	No	No	Yes	No

	StreamingTV	StreamingMovies	Contract	PaperlessBilling
## 1	No	No	Month-to-month	Yes
## 2	No	No	One year	No
## 3	No	No	Month-to-month	Yes
## 4	No	No	One year	No
## 5	No	No	Month-to-month	Yes
## 6	Yes	Yes	Month-to-month	Yes

	PaymentMethod	MonthlyCharges	TotalCharges	Churn	tenure_group
## 1	Electronic check	29.85	29.85	No	0-12 Month
## 2	Mailed check	56.95	1889.50	No	24-48 Month
## 3	Mailed check	53.85	108.15	Yes	0-12 Month
## 4	Bank transfer (automatic)	42.30	1840.75	No	24-48 Month
## 5	Electronic check	70.70	151.65	Yes	0-12 Month
## 6	Electronic check	99.65	820.50	Yes	0-12 Month

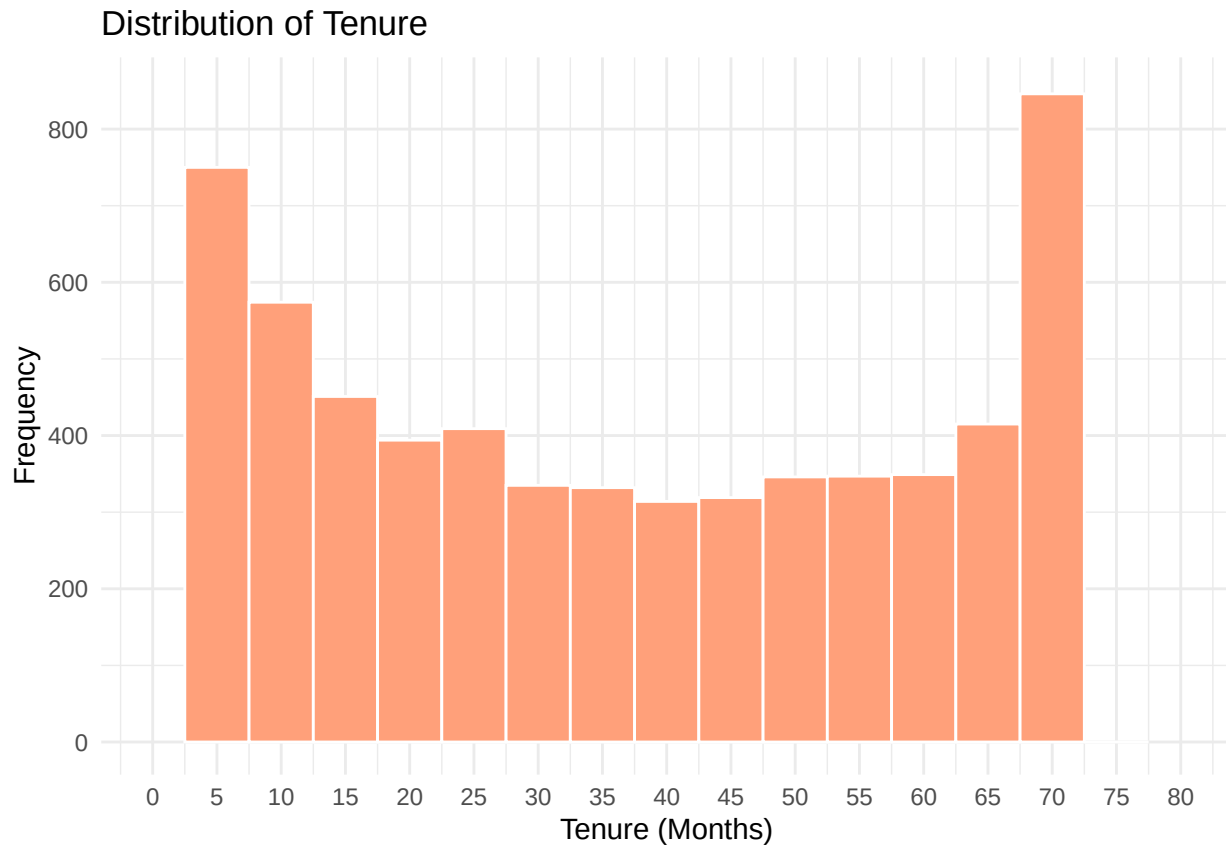
Visualizations

Let's take a look at numerical Features

```
# Histogram of Tenure:
p1 <- ggplot(datrecode, aes(x = tenure)) +
  geom_histogram(binwidth = 5, color = "white", fill = "#FFA07A") +
  labs(title = "Distribution of Tenure",
       x = "Tenure (Months)",
       y = "Frequency") +
  scale_x_continuous(limits = c(0, 80), breaks = seq(0, 80, 5)) +
  theme_minimal()

p1
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```

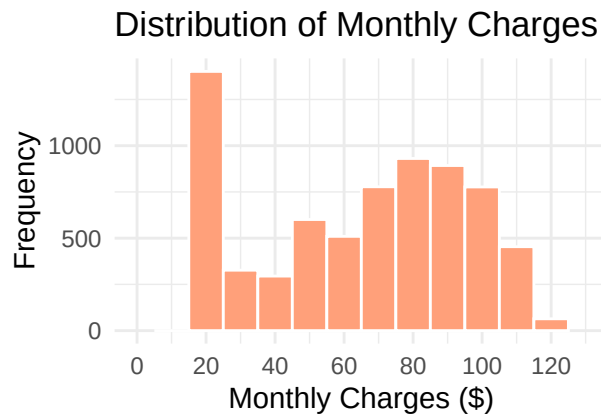
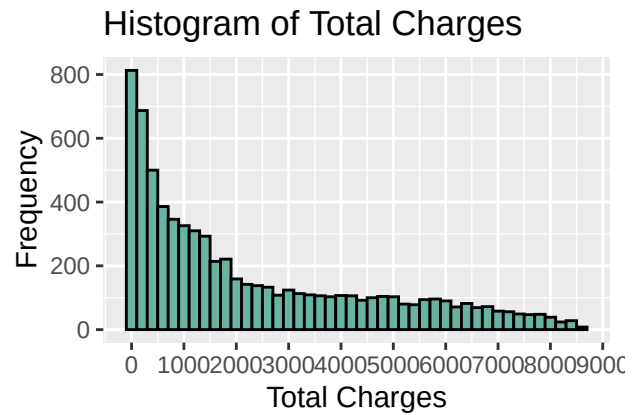
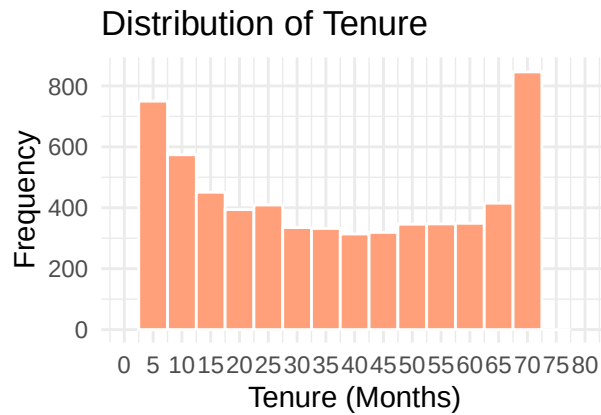


```
# Histogram of TotalCharges
p2 <- ggplot(datrecode, aes(x = TotalCharges)) +
  geom_histogram(binwidth = 200, fill = "#69b3a2", color = "black") +
  scale_x_continuous(breaks = seq(0, 9000, by = 1000)) +
  labs(x = "Total Charges", y = "Frequency",
       title = "Histogram of Total Charges")

# Histogram of MonthlyCharges
p3 <- ggplot(data = datrecode, aes(x = MonthlyCharges)) +
  geom_histogram(binwidth = 10, color = "white", fill = "#FFA07A") +
  labs(title = "Distribution of Monthly Charges",
       x = "Monthly Charges ($)",
       y = "Frequency") +
  scale_x_continuous(limits = c(0, 130), breaks = seq(0, 150, 20)) +
  theme_minimal()

grid.arrange(p1, p2, p3, ncol = 2)
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_bar()`).
## Removed 2 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```



Let's take a look at Categorical Features ## Bar graph of Contract

```
p9 <- ggplot(datrecode, aes(x = Contract, fill = Contract)) +
  geom_bar() + geom_text(aes(y = ..count.. -400,
                             label = paste0(round(prop.table(..count..),
                             1)*100, "%")), stat = "count")+labs(title = "Distribution of Contract Type",
x = "Contract Type",
y = "Count") +
  scale_fill_brewer(palette = "Set2") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Bar graph of gender

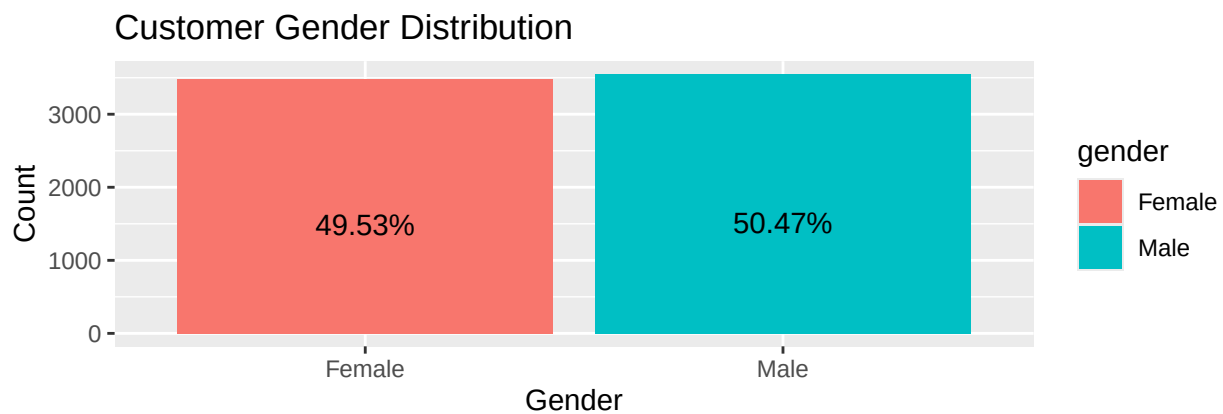
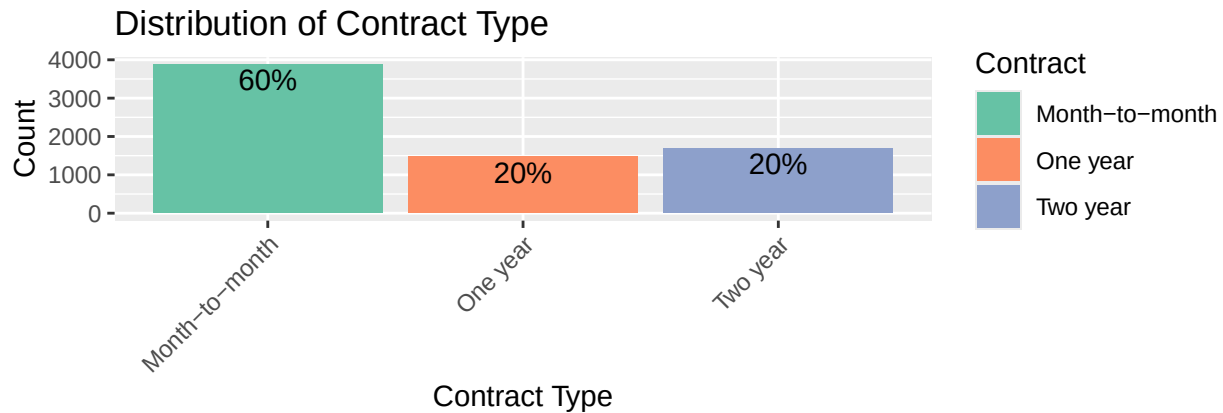
```
colors <- c("#0072B2", "#E69F00")

p10 <- ggplot(data = datrecode, aes(x = gender))+
  geom_bar(aes(fill = gender))+
  geom_text(aes(y = ..count.. -500,
               label = paste0(round(prop.table(..count..),4)* 100, '%')), stat = "count",
position =
  position_stack(vjust = 0.5))+
  xlab("Gender")+
  ylab("Count")+
  ggtitle("Customer Gender Distribution")+
  theme()

# Arrange plots in a grid
```

```
grid.arrange(p9, p10)
```

```
## Warning: The dot-dot notation (`..count..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(count)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

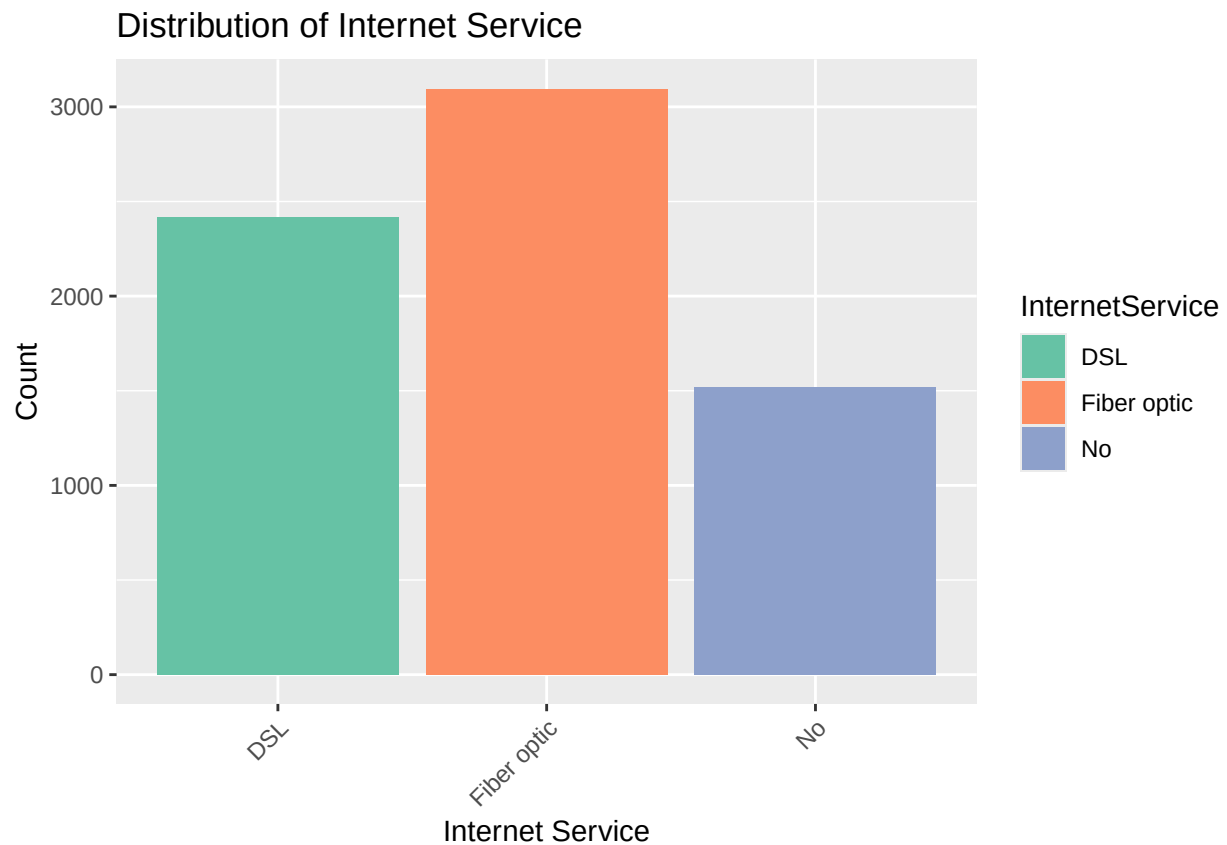


Bar graph of InternetService

```
p11 <- ggplot(datrecode, aes(x = InternetService, fill = InternetService)) +
  geom_bar() +
  labs(title = "Distribution of Internet Service",
       x = "Internet Service",
       y = "Count") +
  scale_fill_brewer(palette = "Set2") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

p11

#

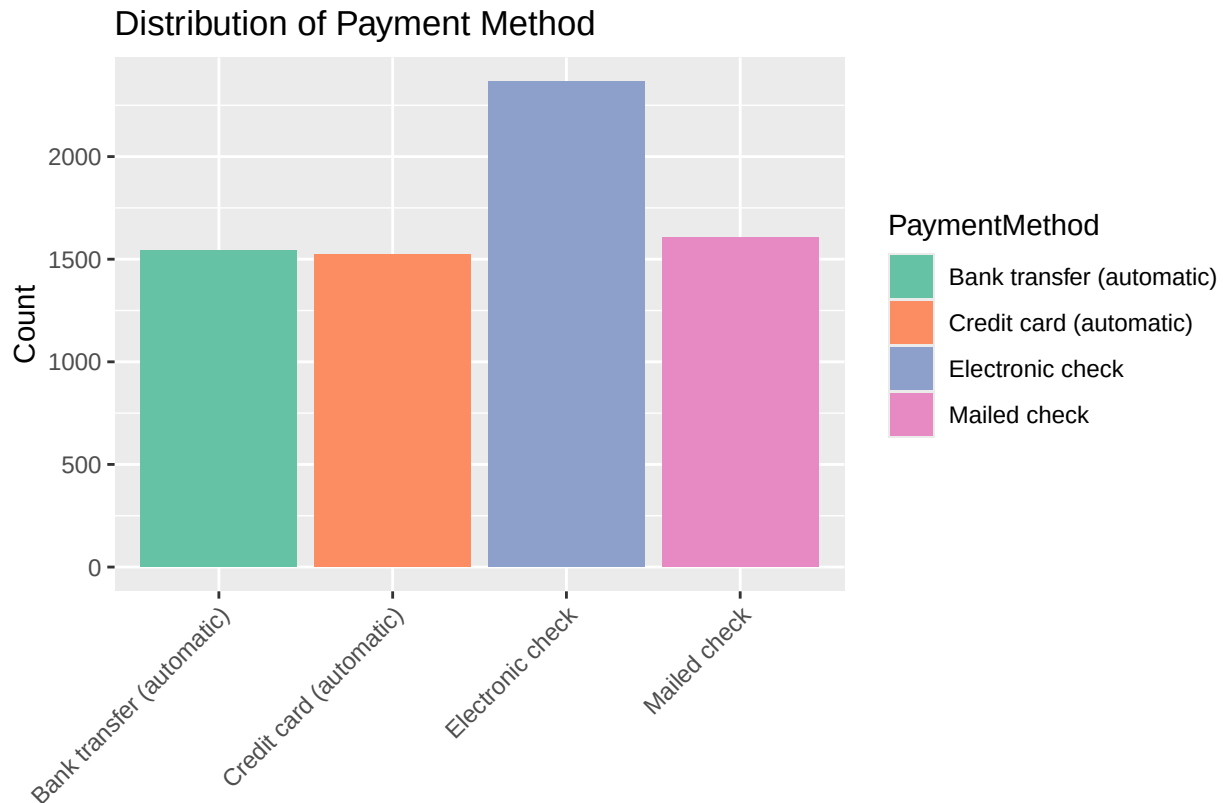


Bar graph of PaymentMethod

```
p12 <- ggplot(datrecode, aes(x = PaymentMethod, fill = PaymentMethod)) +
  geom_bar() +
  labs(title = "Distribution of Payment Method",
        x = "Payment Method",
        y = "Count") +
  scale_fill_brewer(palette = "Set2") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

p12

#



Payment Method

Plotting Bar graphs of SeniorCitizen, Partner, Dependents, PhoneService, # MultipleLines and OnlineSecurity.

```
p14 <- ggplot(datrecode, aes(x = SeniorCitizen)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, '%'),
                stat = 'count')) + ggtitle("SeniorCitizen")

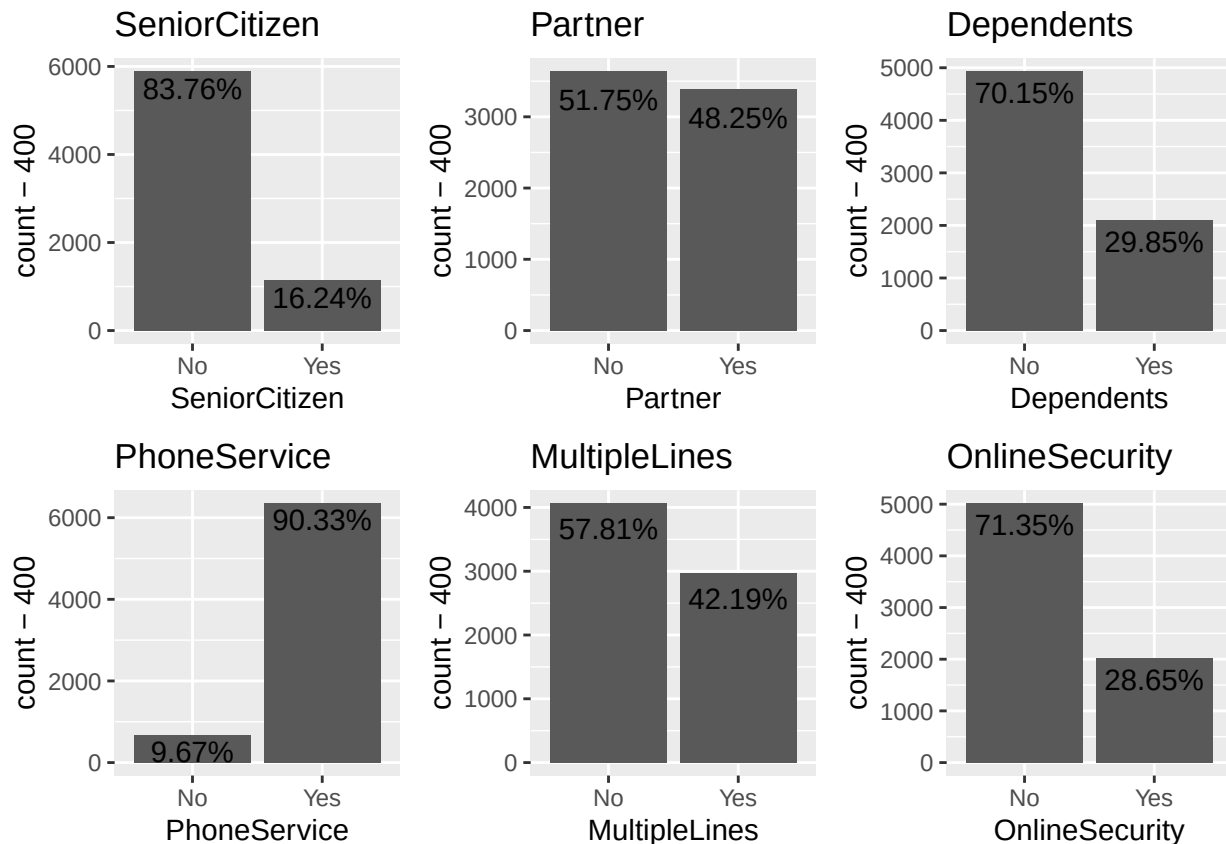
p15 <- ggplot(datrecode, aes(x = Partner)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, '%'),
                stat = 'count')) + ggtitle("Partner")

p16 <- ggplot(datrecode, aes(x = Dependents)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, '%'),
                stat = 'count')) + ggtitle("Dependents")

p17 <- ggplot(datrecode, aes(x = PhoneService)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, '%'),
                stat = 'count')) + ggtitle("PhoneService")

p18 <- ggplot(datrecode, aes(x = MultipleLines)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, '%'),
                stat = 'count')) + ggtitle("MultipleLines")
```

```
p19 <- ggplot(datrecode, aes(x = OnlineSecurity)) + geom_bar() +
  geom_text(aes(y = ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, '%'),
                stat = 'count')+ ggtitle("OnlineSecurity"))
# Arrange plots in a grid
grid.arrange(p14,p15,p16,p17,p18,p19, nrow = 2, ncol = 3)
```



Bar graphs of OnlineBackup, DeviceProtection, TechSupport, StreamingTV, # StreamingMovies and paperLessbill.

```
p20 <- ggplot(datrecode, aes(x = OnlineBackup)) + geom_bar() +
  geom_text(aes(y = ..count.. -1000,
                label = paste0(round(prop.table(..count..),4)*100, '%'),
                stat = 'count')+ ggtitle("OnlineBackup"))

p21 <- ggplot(datrecode, aes(x = DeviceProtection)) + geom_bar()+
  geom_text(aes(y= ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, "%"),
                stat = "count") +ggtitle("DeviceProtection"))

p22 <- ggplot(datrecode, aes(x = TechSupport)) + geom_bar()+
  geom_text(aes(y= ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, "%"),
                stat = "count") +ggtitle("TechSupport"))

p23 <- ggplot(datrecode, aes(x = StreamingTV)) + geom_bar() +
  geom_text(aes(y= ..count.. -400,
                label = paste0(round(prop.table(..count..),4)*100, "%"),
```

```

stat = "count") + ggtitle("StreamingTV")

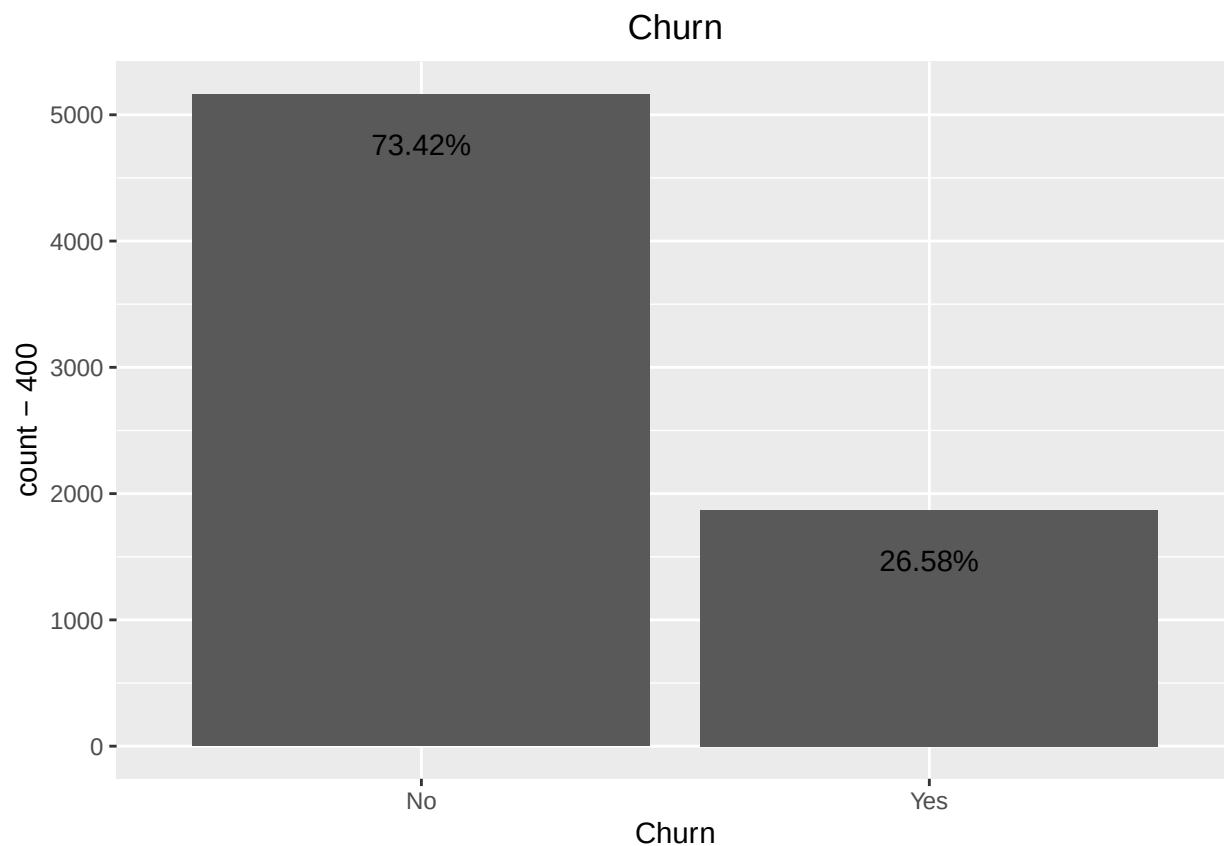
p24 <- ggplot(datrecode, aes(x = StreamingMovies)) +
  geom_bar() + geom_text(aes(y= ..count.. -400,
    label = paste0(round(prop.table(..count..),4)*100, "%")),
    stat = "count") + ggtitle("StreamingMovies")

p25 <- ggplot(datrecode, aes(x = PaperlessBilling)) +
  geom_bar() + geom_text(aes(y= ..count.. -400,
    label = paste0(round(prop.table(..count..),4)*100, "%")),
    stat = "count") + ggtitle("PaperlessBilling")

# How many people has chuned from our data
p13 <- ggplot(datrecode, aes(x = Churn)) +
  geom_bar() + geom_text(aes(y = ..count.. -400,
    label = paste0(round(prop.table(..count..),4)*100, '%')),
    stat = 'count') + ggtitle("Churn") +
  theme(plot.title = element_text(hjust = .5))

p13

```



```

# Calculate average total charges for senior citizens
avg_total_charges_senior <- mean(churn[churn$SeniorCitizen == 1, ]$TotalCharges)

# Calculate average total charges for non-senior citizens

```

```

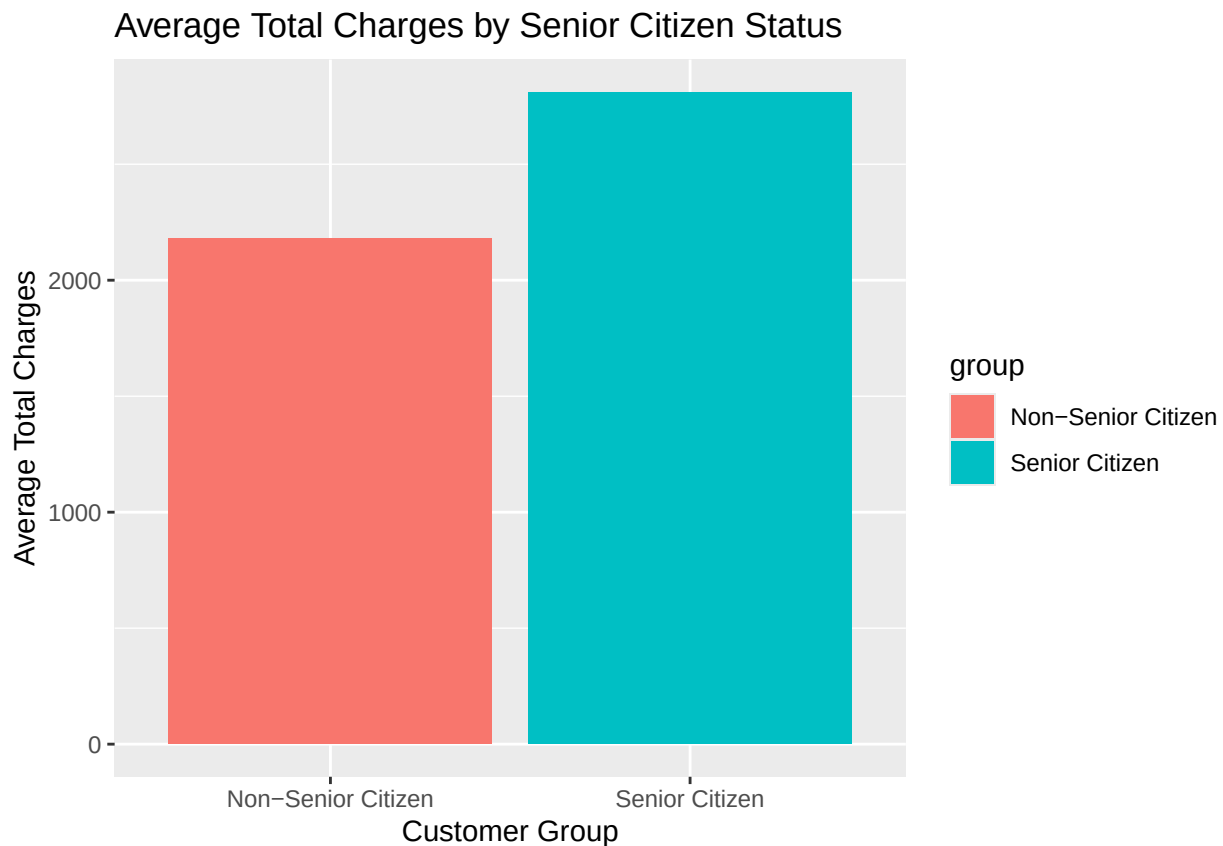
avg_total_charges_non_senior<-mean(churn[churn$SeniorCitizen==0, ]$TotalCharges)

# Create a bar chart for visualization

ggplot(data.frame(group = c("Senior Citizen", "Non-Senior Citizen"),
  avg_charges = c(avg_total_charges_senior, avg_total_charges_non_senior)),
  aes(x = group, y = avg_charges, fill = group)) +
  geom_bar(stat = "identity") +

  labs(title = "Average Total Charges by Senior Citizen Status",
    x = "Customer Group",
    y = "Average Total Charges")

```



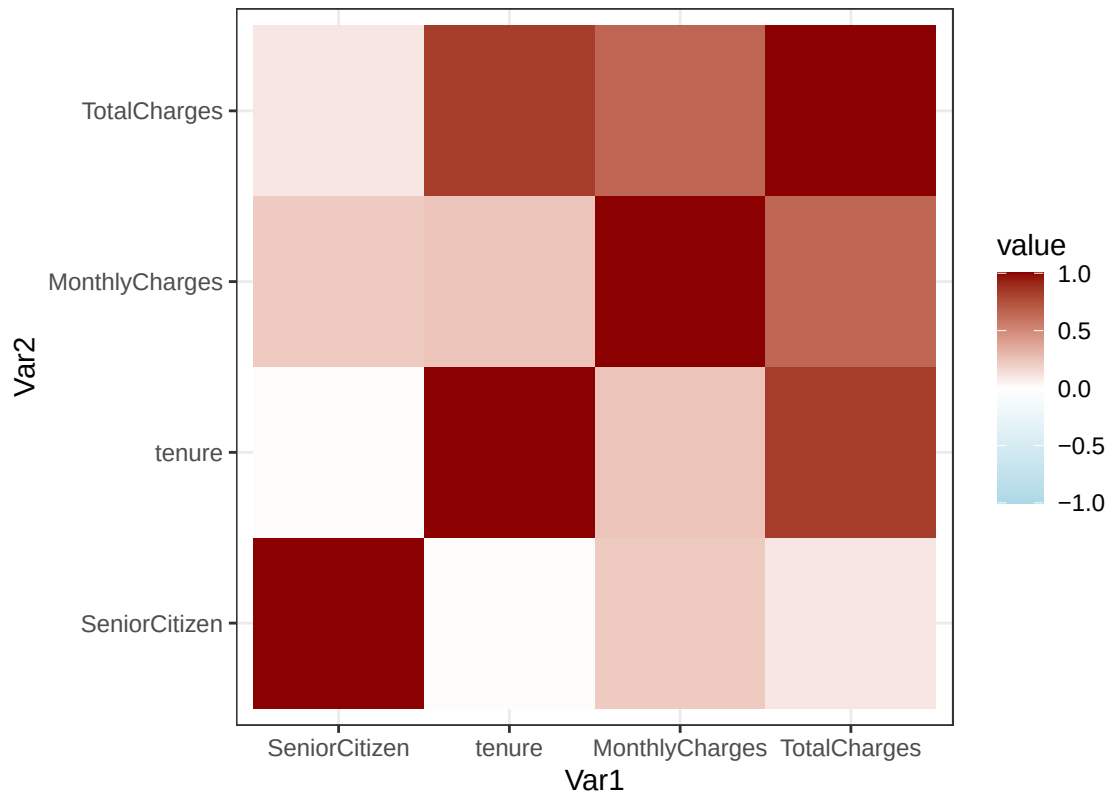
```

correlation_matrix <- cor(churn[, sapply(churn, is.numeric)])

ggplot(melt(correlation_matrix), aes(x = Var1, y = Var2, fill = value)) +
  geom_tile() +
  scale_fill_gradient2(low="lightblue",high="darkred",midpoint=0,limits=c(-1,1)) +
  theme_bw() +
  labs(title = "Correlation Heatmap for Telco Churn Variables") +
  coord_fixed()

```

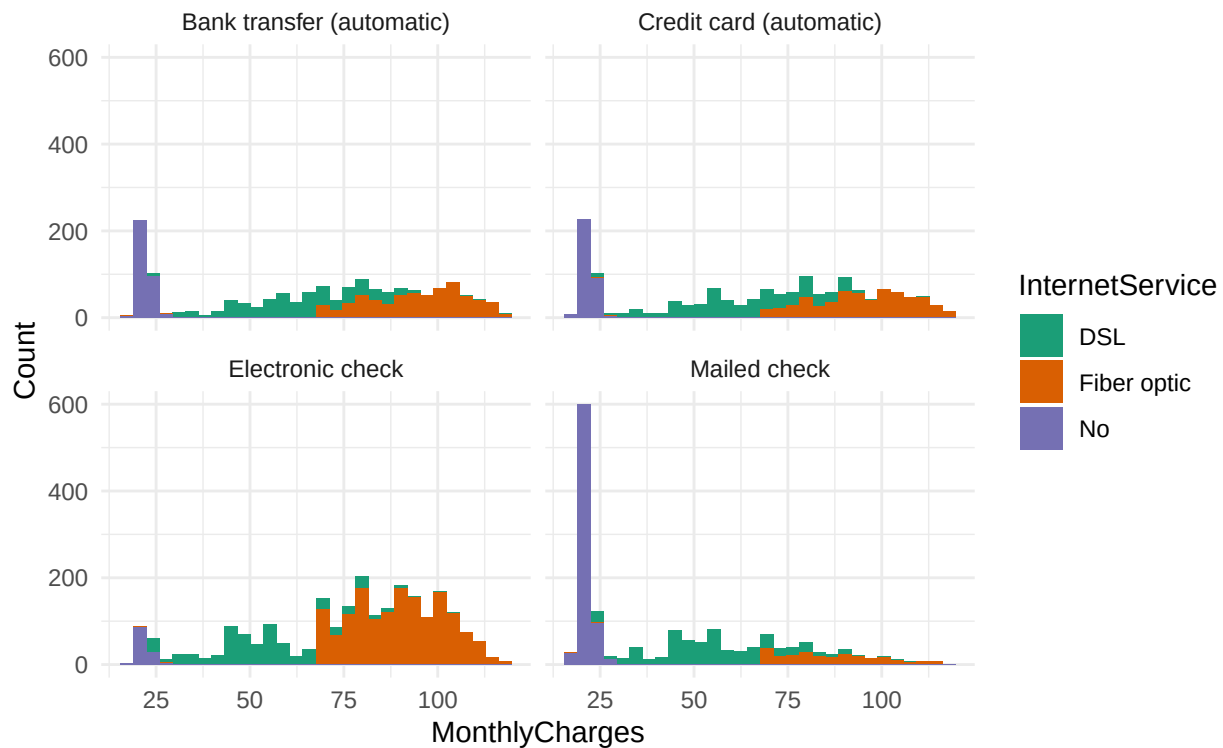
Correlation Heatmap for Telco Churn Variables



```
churn %>% ggplot(aes(x = MonthlyCharges, fill = InternetService)) +
  facet_wrap(~PaymentMethod) + theme_minimal() +
  scale_fill_brewer(palette="Dark2") +
  ggtitle(label = "Monthly Charges paid by customers through various payment
    methods with or without internet") +
  ylab(label = "Count") + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

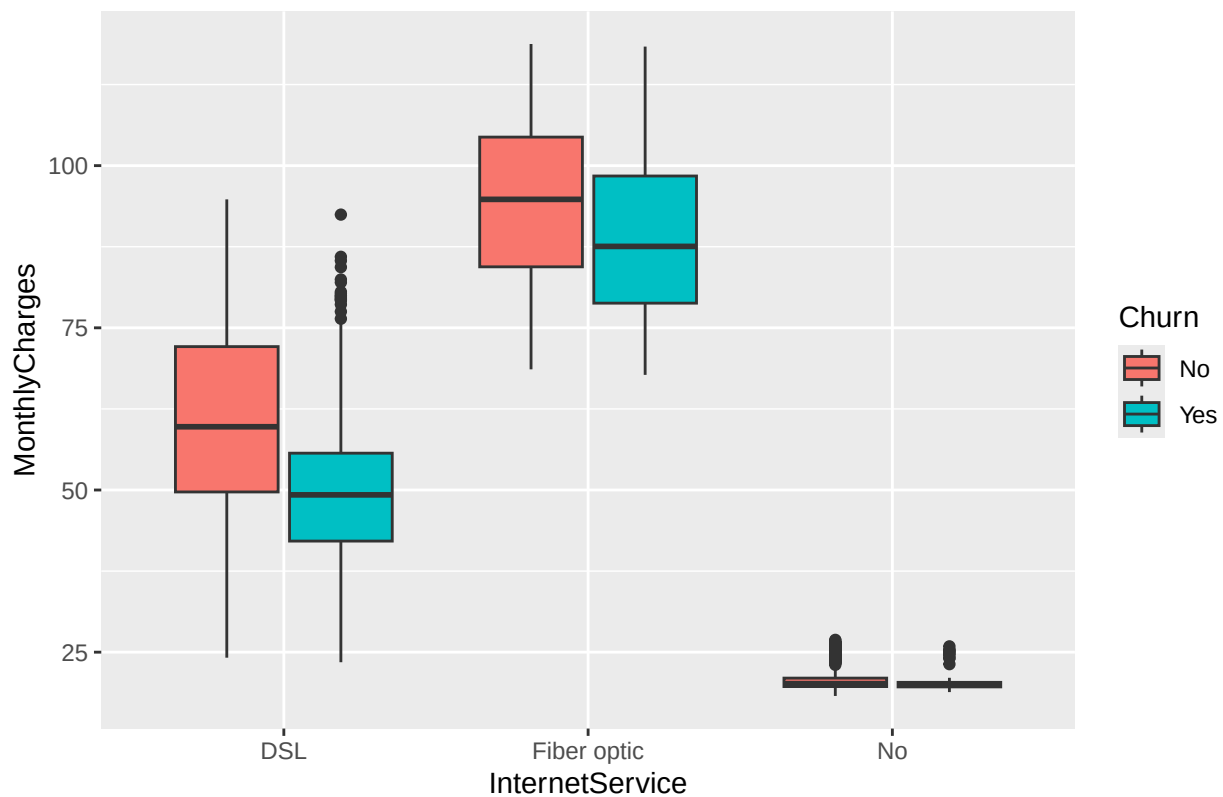
Monthly Charges paid by customers through various payment methods with or without internet



```
# Calculate average monthly charges per internet service type
avg_charges_by_service <- datrecode %>%
  group_by(InternetService) %>%
  summarize(avg_monthly_charge = mean(MonthlyCharges))

# Visualize distribution of average monthly charges by churn status
ggplot(datrecode, aes(x = InternetService, y = MonthlyCharges, fill = Churn)) +
  geom_boxplot() +
  labs(title = "Average Monthly Charges by Internet Service and Churn")
```

Average Monthly Charges by Internet Service and Churn



```
# Correlation analysis between average monthly charges and churn
cor(datrecode$MonthlyCharges, as.numeric(datrecode$Churn))
```

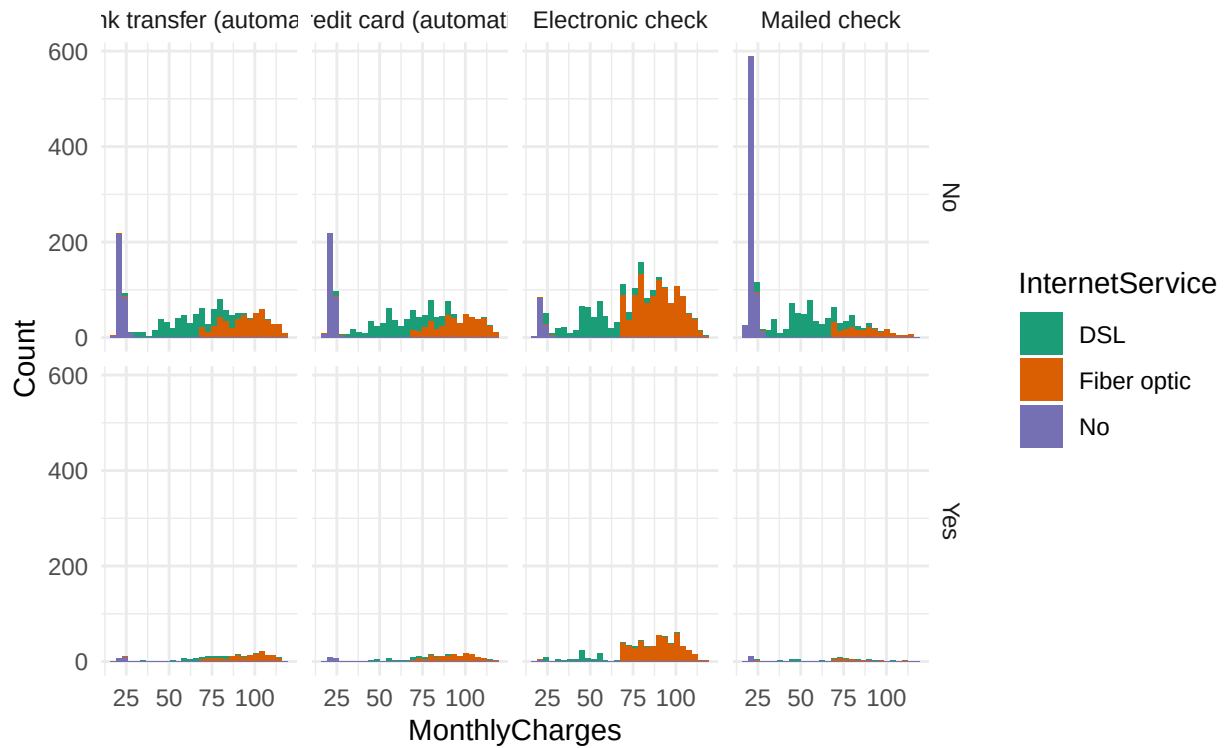
```
## Warning in is.data.frame(y): NAs introduced by coercion
```

```
## [1] NA
```

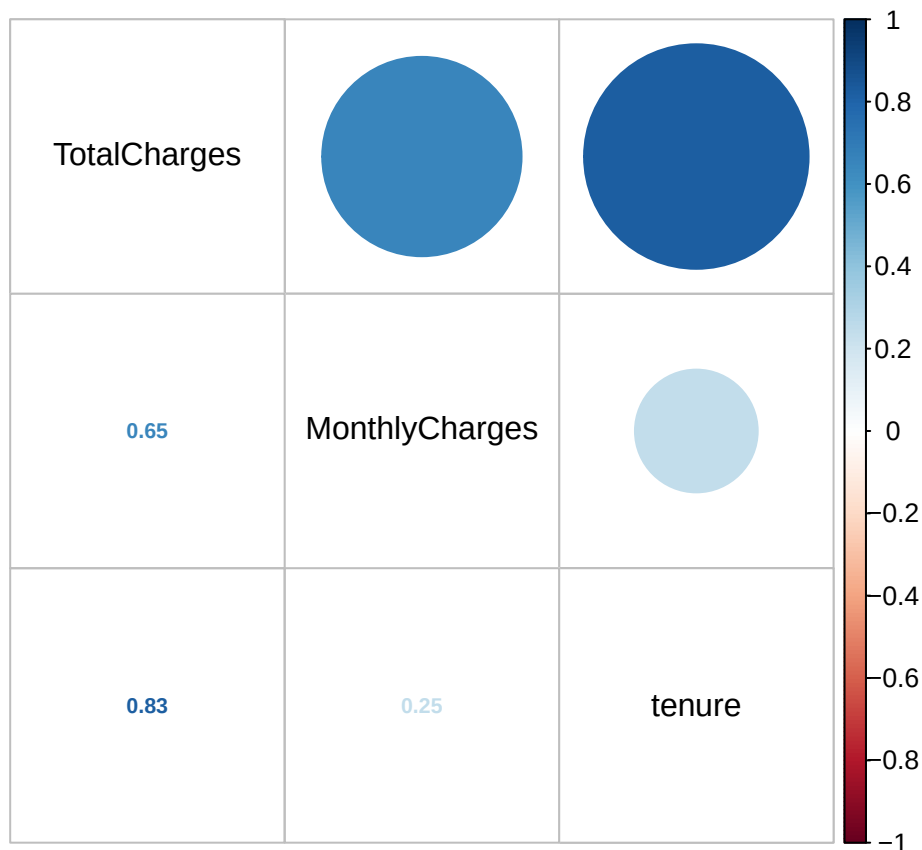
```
# Calculate Distribution of Monthly Charges paid by Senior Citizen Customers
ggplot(datrecode, aes(x = MonthlyCharges, fill = InternetService)) +
  geom_histogram() +
  facet_grid(SeniorCitizen ~ PaymentMethod) +
  theme_minimal() +
  scale_fill_brewer(palette="Dark2") +
  ggtitle(label = "Distribution of Monthly Charges paid by Senior
  Citizen Customers") +
  ylab(label = "Count")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```


Distribution of Monthly Charges paid by Senior Citizen Customers



```
datrecode %>%
  dplyr::select (TotalCharges, MonthlyCharges, tenure) %>%
  cor() %>%
  corrplot.mixed(upper = "circle", tl.col = "black", number.cex = 0.7)
```



```
# recode Churn values from "no/yes" to "0/1"
churn$Churn <- as.factor(mapvalues(churn$Churn,from=c("No", "Yes")
                                   ,to=c("0", "1")))

intrain<- createDataPartition(as.character(churn$Churn),p=0.7,list=FALSE)

training<- churn[intrain,]
testing<- churn[-intrain,]

# Remove Total Charges from the training dataset
dtrain <- intrain[,-19]
dtest  <- intrain[,-19]

dim(training); dim(testing)

## [1] 4924  21
## [1] 2108  21

lr_fit <- glm(Churn ~.,data = training,family=binomial(link='logit'))
summary(lr_fit)

##
## Call:
## glm(formula = Churn ~ ., family = binomial(link = "logit"), data = training)
##
## Coefficients:
##                                     Estimate Std. Error z value Pr(>|z|)
```

```

## (Intercept)                3.815e+00  1.156e+00   3.300 0.000968 ***
## genderMale                 -3.617e-02  7.821e-02  -0.462 0.643797
## SeniorCitizen              2.903e-01  1.014e-01   2.863 0.004194 **
## PartnerYes                 8.589e-02  9.446e-02   0.909 0.363174
## DependentsYes              -1.717e-01  1.088e-01  -1.578 0.114660
## tenure                     -8.071e-02  1.162e-02  -6.944 3.82e-12 ***
## PhoneServiceYes            8.794e-01  7.808e-01   1.126 0.259998
## MultipleLinesYes           7.467e-01  2.140e-01   3.490 0.000483 ***
## InternetServiceFiber optic  2.782e+00  9.628e-01   2.890 0.003854 **
## InternetServiceNo          -2.581e+00  9.692e-01  -2.663 0.007750 **
## OnlineSecurityYes          4.659e-02  2.142e-01   0.217 0.827820
## OnlineBackupYes            3.344e-01  2.108e-01   1.587 0.112612
## DeviceProtectionYes        3.005e-01  2.118e-01   1.418 0.156059
## TechSupportYes             1.284e-01  2.178e-01   0.590 0.555440
## StreamingTVYes             1.009e+00  3.941e-01   2.559 0.010491 *
## StreamingMoviesYes         9.721e-01  3.940e-01   2.467 0.013627 *
## ContractOne year           -6.641e-01  1.311e-01  -5.064 4.10e-07 ***
## ContractTwo year           -1.491e+00  2.203e-01  -6.770 1.29e-11 ***
## PaperlessBillingYes        3.473e-01  9.024e-02   3.848 0.000119 ***
## PaymentMethodCredit card (automatic) -5.893e-02  1.362e-01  -0.433 0.665155
## PaymentMethodElectronic check 2.829e-01  1.144e-01   2.472 0.013432 *
## PaymentMethodMailed check  -1.110e-01  1.401e-01  -0.792 0.428236
## MonthlyCharges             -7.831e-02  3.820e-02  -2.050 0.040363 *
## TotalCharges               2.513e-04  9.179e-05   2.738 0.006177 **
## tenure_group0-12 Month     -1.566e+00  5.792e-01  -2.704 0.006854 **
## tenure_group12-24 Month    -1.697e+00  4.757e-01  -3.568 0.000360 ***
## tenure_group24-48 Month    -1.081e+00  3.392e-01  -3.186 0.001443 **
## tenure_group48-60 Month    -3.415e-01  2.282e-01  -1.496 0.134572
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 5702.8 on 4923 degrees of freedom
## Residual deviance: 4028.0 on 4896 degrees of freedom
## AIC: 4084
##
## Number of Fisher Scoring iterations: 6
anova(lr_fit, test="Chisq")

## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: Churn
##
## Terms added sequentially (first to last)
##
##
##              Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL              4923      5702.8
## gender             1      0.72      4922      5702.0 0.3959161
## SeniorCitizen      1    113.69      4921      5588.4 < 2.2e-16 ***
## Partner            1     98.80      4920      5489.6 < 2.2e-16 ***

```

```
## Dependents      1      30.67      4919      5458.9 3.052e-08 ***
## tenure          1      643.43      4918      4815.4 < 2.2e-16 ***
## PhoneService    1         0.61      4917      4814.8 0.4341878
## MultipleLines   1      140.08      4916      4674.8 < 2.2e-16 ***
## InternetService 2      425.47      4914      4249.3 < 2.2e-16 ***
## OnlineSecurity  1       26.01      4913      4223.3 3.405e-07 ***
## OnlineBackup    1         0.04      4912      4223.2 0.8422295
## DeviceProtection 1         1.03      4911      4222.2 0.3112378
## TechSupport     1       14.20      4910      4208.0 0.0001648 ***
## StreamingTV     1       18.88      4909      4189.1 1.392e-05 ***
## StreamingMovies 1         7.74      4908      4181.4 0.0053916 **
## Contract        2       67.48      4906      4113.9 2.226e-15 ***
## PaperlessBilling 1       16.30      4905      4097.6 5.419e-05 ***
## PaymentMethod   3       16.26      4902      4081.4 0.0010052 **
## MonthlyCharges  1         3.38      4901      4078.0 0.0658483 .
## TotalCharges    1       24.71      4900      4053.3 6.677e-07 ***
## tenure_group    4       25.31      4896      4028.0 4.365e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Assessing the predictive ability of the Logistic Regression model
```

```
testing$Churn <- as.character(testing$Churn)
testing$Churn[testing$Churn=="No"] <- "0"
testing$Churn[testing$Churn=="Yes"] <- "1"
fitted.results <- predict(lr_fit,newdata=testing,type='response')

misClasificError <- mean(fitted.results != testing$Churn)
print(paste('Logistic Regression Accuracy',1-misClasificError))
```

```
## [1] "Logistic Regression Accuracy 0"
```

```
print("Confusion Matrix for Logistic Regression");
```

```
## [1] "Confusion Matrix for Logistic Regression"
```

```
table(testing$Churn,fitted.results > 0.5)
```

```
##
##      FALSE TRUE
## 0   1393   155
## 1    266   294
```

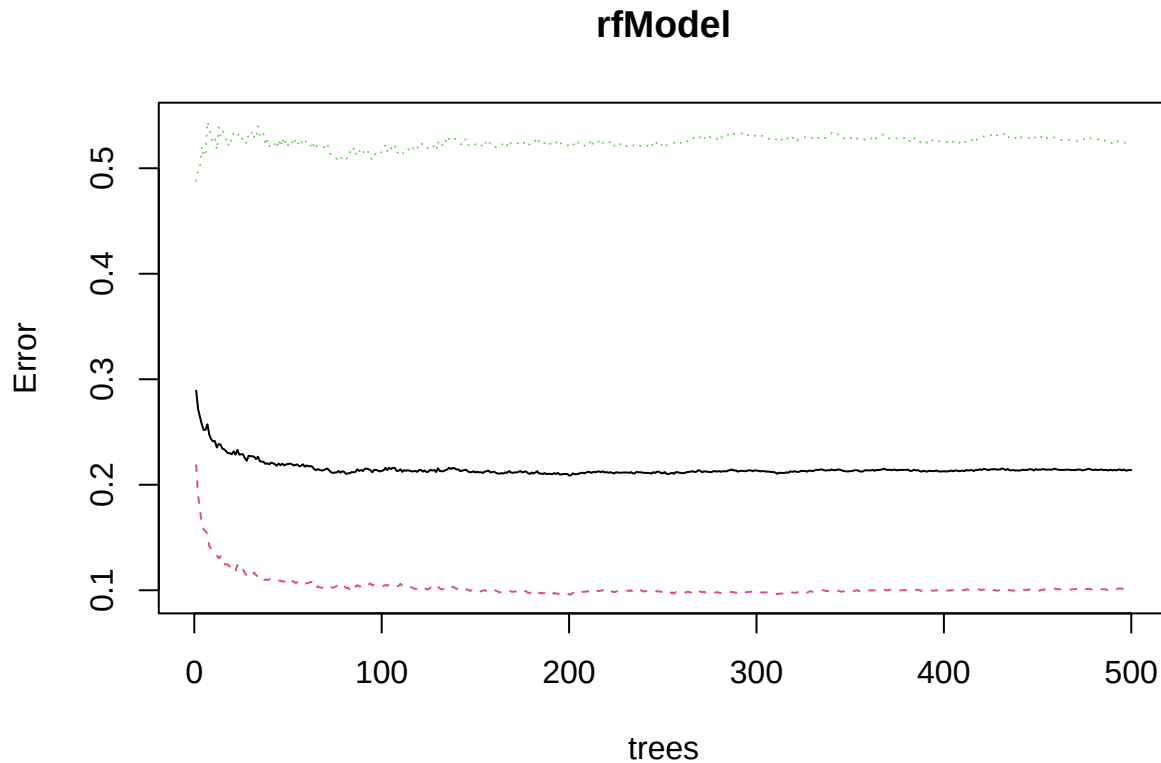
```
# Decision Tree Visualization
```

```
rfModel <- randomForest(Churn ~.,data = training)
print(rfModel)
```

```
##
## Call:
## randomForest(formula = Churn ~ ., data = training)
##              Type of random forest: classification
##              Number of trees: 500
## No. of variables tried at each split: 4
##
##              OOB estimate of  error rate: 21.39%
## Confusion matrix:
##           0    1 class.error
## 0 3251 364    0.1006916
```

```
## 1 689 620 0.5263560
```

```
plot(rfModel)
```



```
rfModel_new <- randomForest(Churn ~ ., data = training, ntree = 200, mtry = 2
                             , importance = TRUE, proximity = TRUE)
print(rfModel_new)
```

```
##
## Call:
## randomForest(formula = Churn ~ ., data = training, ntree = 200,      mtry = 2, importance = TRUE, p
##               Type of random forest: classification
##               Number of trees: 200
## No. of variables tried at each split: 2
##
##           OOB estimate of  error rate: 20.41%
## Confusion matrix:
##      0   1 class.error
## 0 3300 315  0.08713693
## 1  690 619  0.52711994
```

```
# Random Forest Feature Importance
varImpPlot(rfModel_new, sort=T, n.var = 10, main='Top 10 Feature Importance')
```

Top 10 Feature Importance

